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BIOLOGICAL PROPERTIES OF SOME MEDICALLY IMPORTANT SNAKE SPECIES OF PAKISTAN

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ABSTRACT

Venoms from eight medically important snake species including three sea snake species, belonging to families Viperidae and Elapidae, were tested for lethal, hemolytic (direct and indirect) edema-inducing and hemorrhagic activities. The species includes five from family Elapidae; *Hydrophis cyanocinctus, Enhydrina schistosa, Naja naja naja. Bungarus caeruleus* and *Microcephalophis gracilis gracilis* whereas three from family Viperidae; *Vipera russelli russelli, Eristochphis macmahoni* and *Echis carinatus*. All venom tested moderate to high levels of lethal, hemolytic (both direct and indirect) and edema-inducing activities, except that venoms of snake species of family Elapidae, including the sea snake species, are characterized by low levels or absence of hemorrhagic activity. LD_{50} values were found to be 0.39 mg/kg 0.41 mg/kg 0.28 mg/kg. 0.30 mg/kg and 0.40 mg/kg for the species of family Elapidae respectively whereas 0.30 mg/kg, 0.26 mg/kg and 0.41 mg/kg for species of family Viperidae.

INTRODUCTION

Venomous snakes are generally divided into three families; Colubridae, Viperidae and Elapidae (Tan, 1991). Of these three, species belonging to families Viperidae and Elapidae are also found abundantly in Pakistan and considered as medically important. Family Viperidae consist of land snake only whereas Elapidae also represent sea snakes of subfamilies *Laticaudinae* and *Hydrophiinae*.

Snakes of both families are known to cause oaralysis, respiratory failure, neurological disorders, hemorrhagic, edema and bleeding (Minton, 1974; Russell, 1984). Besides these symptoms, muscular pain, myoglobinuria and skeletal muscle necrosis have also been reported (Minton, 1974).

In this communication we report the biological properties of venoms from eight medically important snake species belonging to families Viperidae and Elapidae including three sea snake species. The species includes five from family Elapidae; *Hydrophis cyanocinctus, Enhydrina schistosa, Naja naja naja, Bungarus caeruleus* and *Microcephalophis gracilis gracilis* whereas three from family Viperidae; *Vipera russelli russelli. Eristocophis macmahoni* and *Echis carinatus*. The observation from the study will be helpful in the evaluation of clinical manifestations produced by venom components and its prevention.

MATERIALS AND METHODS

Materials

The sea snake species were captured near and around coastal areas of Karachi whereas other specimen were supplied by the serpent charmers and identified by Zoological Survey department. Venom was collected manually by milking technique from live snakes and lyophilized.

Lethality Determination

Group of six albino rats (100-150 gm) of either sex were injected (i.p.) with various amounts of each venom dissolved in 0.5 ml phosphate buffered saline, PBS (pH 7.0). Deaths were observed at 4 h and LD_{50} determined by plotting log dose versus percent death.

Edema-inducing activity

Edema-inducing activity was determined by the method described by Tan and Saifuddin (1990). The unit of edema ratio, is defined as the percentage of the weight of the edematous leg to the weight of the normal leg (control). Minimum edema dose (MED) is defined as the amount of substance causing a edema ratio of 130% by plotting edema ratio versus log dose of substance.

Assay of direct hemolytic activity

The direct hemolytic activity was assayed by the method described by Yang and Lee (1978) using 1% guinea pig RBCs suspension. Known quantity of venom solution was incubated with 1% RBC suspension and hemolysis was measured at 550 nm. Result is expressed as percent hemolysis.

Assay of indirect hemolytic activity

Indirect hemolytic activity was determined by pre-incubation of lecithin with known quantity of venom solution as described by Yang and Lee (1978). After incubation the lecithin and venom suspension was treated similarly as in assay for direct hemolvtic activity. Result is expressed as percent hemolysis.

Hemorrhage activity

Hemorrhage activity was estimated in rats and mice according to the method of Ferreira *et al.*, (1992) using 50 µl venom solution in saline. Venom solution was injected i.d. After 2 hr. the animals were killed with ether and hemorrhage was measured by mm. scale. The minimum hemorrhage dose (MHD) was defined as the least amount of venom causing a hemorrhage reaction of 5mm in diameter after 2 hr.

RESULTS AND DISCUSSION

Results are summarized in Table 1.

Lethality

The LD_{50} was found to be in the range of 0.26 mg/kg to 0.41 mg/kg by i.p. route. The envenomated rats showed severe ataxia, paresis, bradycardia, labored breathing and respiratory arrest. Bleeding from nostrils and jaws were seen in case of rats given *V. russelli* and *E. macmahoni venom*. In those instances where 100% lethality was present, chronic convulsion was observed before death. Our results are in good agreement with LD_{50}s reported for sea snakes. *H. cyanocinctus*, *Enhydrina schistosa* and *L. hardwickii* (Limpus, 1978; Yang and Lee, 1978) from other parts of the world. The gross action of the venom observed in present study are typical of Viperidae and Elapidae venom and also reported for sea snakes *Aipusurus laevis*. *Astrlia stokerii*, *Hydrophis elegans* (Limpus, 1978), *Cerastes cerastes* and *Bungarus fasciatus* (Tan, 1991).

Hemolytic activity

No direct hemollytic activity was observed in RBC suspension upto the concentration of 180 µg/ml RBC suspension in case of sea snake venoms whereas same venoms induced marked indirect hemolytic activity. The results clearly indicate that although sea snake venom are basically non-hemolytic but possess considerable phospholipase A (PLA) activity (Yang and Lee. 1978). PLA has long been associated with the hemolytic activity of snake venoms. Its hemolytic activity *in vitro* depends upon its association with other factor as well (Condrea and de Varies. 1965). Absence of direct hemolytic activity is also a characteristic of sea-snake venom having no apparent effect on blood component and circulating system. Similar results were reported previously for *Hydrophis cyanocinctus* upto a venom concentration of 1×10^{-4} gm/ml (Yang and Lee, 1978). The land snakes, on the other hand, induced moderate to high levels of direct hemolytic activity. This suggest a direct or indirect action of their venom on blood components, since interference with normal blood mechanism is reported to be a property of many snake venoms including *Cerastes cerastes*. *Naja nigricollis* and *Crotalus adamanteus* (Minton, 1974; Mohamed *et al.*, 1969a b.). In addition, systemic bleeding by defibrination and thrombocytopenia have also been reported in pit vipers (Reid *et al.*, 1963).

Edema-inducing activity

Minimum edema dose (MED) was determined by plotting percent edema ratio versus log dose and found to be in the range of 1.62 μg/paw to 10.15 μg/paw. The variation from lowest to highest MED was approximately 7.0-fold. The edema-inducing activity demonstrated the presence of one or several inflammatory agents in these venom which is also a common feature amongst snake venom. The presence of edema-inducing property have also been studied previously in a number of Elapidae and Viperidae venoms (Tan and Saifuddin, 1990) and found to variable amongst genera and even at species level.

Hemorrhage activity

All snake venoms exhibited moderate levels of hemorrhage activity except sea snakes, even at a concentration of 200 μg. Absence of hemorrhage activity was also reported for other sea snake species of genus *Hydrophis, Lapemis, Lacticauda* and *Enhydrina* species (Tan and Ponnudurai, 1991). It is reported that venom of sea snakes of both sub-families, Hydrophiinae and Lacticaudinae are characterized by absence of certain biological activities (Tan and Ponnudurai, 1991). Low hemorrhagic activity of *N. naja naja* and *B. caeruleus* venoms observed in present study were also reported to be a characteristic of Elapidae venom (Tan and Ponnudurai, 1991). Previously it was reported that venoms of *vipera ruselli siamensis, Pseudocerastes persicus fieldi* and *Crotalus durissus terrificus* with little or no hemorrhagic activity were found to be more toxic (Minton, 1974). In our results also both *Naja* and *B. caeruleus* venoms are comparatively less hemorrhagic and thus more lethal than others.

In conclusion, eight snake venoms samples from two families Elapidae and Viperidae, both medically important, have been investigated for their biological properties and found to possess variable degree of moderate to high hemolytic and edema-inducing activities. Elapidae venoms are characterized either by the absence or low levels of direct hemolytic and hemorrhagic activities.

Table 1 Biological Activities of Snake Venoms from Families Viperidae and Elapidae

Species	Lethality LD_{50} i.p.	H. Activ.		Edema-inducing Activity µg/paw (MED)	Hemorr-hagic Activity µg/mouse (MHD)
		Direct %	Indirect		
Elapidae					
H. Cyan	0.39 ±0.006	7.65 ±0.37	75.00 ±2.04	1.62 ±0.04	—
E. Schi	0.41 ±0.008	8.62 ±0.55	70.25 ±0.03	2.50 ±0.04	—
M. GG.	0.40 ±0.006	8.87 ±0.42	79.25 ±1.49	2.00 ±0.04	- —
N. Naja	0.28 ±0.010	84.75 ±1.88	65.75 ±2.17	2.95 ±0.06	61.25 ±0.75
B. Caer	0.30 ±0.006	72.25 ±0.85	66.25 ±1.03	10.12 ±0.31	74.25 ±1.10
Viperidae					
V. RR	0.30 ±0.006	70.25 ±0.63	50.25 ±0.85	2.07 ±0.03	54.75 ±1.70
E. Car	0.41 ±0.004	80.00 ±2.04	40.00 ±0.70	4.17 ±0.12	36.50 ±1.20
E. Mach	0.26 ±0.006	66.50 ±0.64	70.25 ±0.85	4.60 ±0.09	42.75 ±1.10

Results are expressed as mean ± S.E.

H. CYAN = *H. cyanocinctus*, *E. SCHI* = *E. schistosa*, *M. GG.* = *M. gracilis gracilis*.
N. NAJA = *N. naja naja*, *B. CAER* = *B. coeruleus*, *V. RR* = *V. russelli russelli*.
E. CAR = *E. carinatus*, *E. MACH* = *E. macmahoni*

H. ACTIV. = HEMOLYTIC ACTIVITY.

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